

Slides 4.12: Plan: Scale Drawing and Traffic Flow

Slide outline for Lesson 4.12. Day 1 has eleven slides; Day 2 has eleven. Keep the conversion line on the board both days: at 1/2 inch equals 1 foot, one square is 6 inches, 6 squares is 36 inches, 5 squares is 30 inches, 10 squares is 60 inches.

Day 1

Slide 1: Do now

- Here is a 12 by 14 room with the bed in the middle of the floor
- Redraw it at 1/4 inch equals 1 foot with the bed against a wall
- Add a straight path from the door to the window
- Two minutes to draw, one to compare with a neighbor Image: a rough plan of a 12 by 14 room with a bed floating in the center of the floor and nothing else. Notes: Day 6 of the floor plan stack. Skim the stack before class for scale errors; if several students drew a 12-foot wall as 12 squares at the wrong scale, slow the demo down.

Slide 2: Thirty inches

- I walk through the tape strip. Easy.
- I carry a laundry basket through it. Harder.
- A student walks through it with the walker. Watch.
- Thirty inches is fine for you. It is not fine for Grandpa Okafor. Image: two labeled masking tape strips on the floor, one 36 inches and one 30 inches apart. Notes: *[Sal: borrow a walker, or a wheelchair from the nurse's office. Ten seconds of rolling through the 30-inch strip does more than any slide.]* Land it: today every path in your plan gets measured.

Slide 3: Plan: Scale Drawing and Traffic Flow

- Lesson 4.12, Day 1 of 2
- Today: draw the room to scale and place the furniture
- Tomorrow: measure every path, get reviewed, revise
- A floor plan is a promise that everything fits and everyone can move Image: a finished student-style bedroom plan with furniture, shaded paths, and the scale written in the corner.

Slide 4: The project scale

- 1/2 inch equals 1 foot. Two graph squares per foot.
- A 12-foot wall is 24 squares
- Tiny house teams: 1/4 inch equals 1 foot on letter paper, or 1/2 inch on 11 by 17

- Write the scale in the bottom corner. Write your client's name at the top. Notes: This is the Lesson 4.10 scale doubled, so a bedroom fills the sheet. Say the rule out loud: the numbers on the card are not up for debate.

Slide 5: Draw the outline

- Read the room measurements on your card
- Count the squares before you draw a line
- Four walls with a ruler
- Check with a partner: does 12 feet equal 24 squares? Image: the Okafor room outline, 11 by 12, drawn on graph paper with the square count marked on two walls. Notes: Draw the Okafor room on the board while they draw theirs. Walk and count squares with them.

Slide 6: Doors, windows, closet

- Door: leave a gap as wide as the door (3 feet is 6 squares), draw the line plus the quarter-circle swing
- Bedroom doors and front doors swing in. The tiny house door swings out.
- Window: a break in the wall with two thin parallel lines
- Closet: a small rectangle the size the card gives, open to the room Image: the three symbols drawn large and labeled: door with swing arc, window, closet. Notes: Common mistake is the swing drawn into the hall. Say it here and say it again on Day 2.

Slide 7: Zones first

- Sleep, work, store, sit, move
- Tiny house: cook, wash, eat, sleep, sit, store, move
- The zone that needs light goes near the window
- The zone that needs quiet goes away from the door
- Pencil each zone as a light circle. Zones are not furniture yet. Image: the Okafor plan with four light circles labeled sleep, sit, store, move.

Slide 8: The furniture cut-outs

- Page 2 of the handout: real furniture sizes at our scale
- A twin bed is 3 by 6.5 feet, which is 6 by 13 squares. A desk is 2 by 4.
- Cut the pieces your client needs. Needs first, wants second.
- Slide them around. Do not glue anything today.
- The bed goes down first, then the chair or desk, then storage Image: the cut-out sheet with a twin bed, a dresser, an armchair, and a nightstand cut out and sitting loose on a plan. Notes: Seventeen minutes on this step. Ask every pair the same question: which piece did you put down first, and why?

Slide 9: Anything not on the sheet

- Measure a real one, or look it up
- Round to the nearest half foot

- Draw the rectangle and write the real size on the piece Notes: This is where the optional homework from Lesson 4.11 pays off. Any student who measured a bed or a dresser at home uses their number.

Slide 10: Freeze the plan

- Trace lightly around every piece, or tape it with removable tape
- Plan and loose pieces go in your project folder
- Tomorrow every path gets measured and a peer gets to find your mistakes Notes: Five minutes. Count the folders as they go in the bin.

Slide 11: Sticky note out

- The first piece I placed was ___ because ___
- The piece I could not fit is ___ Notes: Two read aloud. Stick the notes on the board under each client's name. The "could not fit" notes are your board examples for tomorrow.

Day 2

Slide 12: Do now

- Sketch a 6 by 9 foot entry with the front door on a short wall
- Add a bench and a wall of hooks
- Shade the walking path from the front door to the inside doorway
- Is the path 36 inches? Write yes or no and how you know.
- Two minutes to draw, one to check Notes: Day 7 of the floor plan stack. This is the Nguyen card, so the mudroom students get a free rehearsal. The bench is the usual culprit: an 18-inch bench in a 6-foot wall leaves 54 inches, which passes.

Slide 13: Traffic flow

- Lesson 4.12, Day 2
- Today: measure every path, check the principles, peer review, revise
- A path is as wide as its narrowest point. Measure the pinch, not the entrance. Image: one plan with a path shaded from the door to the bed, wide at the door and pinched to 24 inches at the foot of the bed, with the pinch circled.

Slide 14: The rules

- 36 inches for any walker or wheelchair path, and for a front entry: 6 squares
- 30 inches everywhere else: 5 squares
- A wheelchair turning circle is 60 inches across: a circle 10 squares wide
- The door swing stays clear. Nothing tall in front of a window that must open.
- A bed needs one long side open to a path
- A crib stays 36 inches from a window, cord, heater, or anything that could fall Image: page 3 of the handout with a small picture beside each rule. Notes: These numbers

come from the ADA for the 36-inch and 60-inch lines and from common residential planning for the 30-inch line. A home is not a public building, so this is not code; we use them because they work.

Slide 15: Shade and measure

- Shade every path a person walks: door to each zone, and zone to zone
- Measure the narrowest spot in squares, write the width in inches beside it
- Circle any path that fails
- Draw the turning circle if your client uses a wheelchair
- Eight minutes Notes: Watch for measuring in squares and forgetting to convert. Six squares is 3 feet is 36 inches. Point at the board line every time.

Slide 16: Now check the principles

- Is there a focal point? What does a person see first from the door?
- Is the room balanced, or is everything on one wall?
- Is each piece in proportion to the room? A full bed in an 8 by 9 room eats it.
- Name your focal point on the plan Notes: This is Lessons 4.6 and 4.7 coming back. A plan that passes every path and has nothing to look at is not finished.

Slide 17: Peer review, 10 minutes

- Swap plans with a pair who has a different client
- Use the checklist, page 4. Fifteen boxes, yes or no. Every no gets a note.
- Mark, do not fix. Colored pen. The owner decides.
- Write two things that work for this client and one thing to change Image: the checklist with the first five boxes filled in and a note beside a no. Notes: This is the same routine as Lesson 1.19. A review with fifteen yeses and no notes goes back. "Add more stuff" is not a client need; send the reviewer to the card.

Slide 18: Questions I will ask while you review

- How many squares is 36 inches at our scale? At 1/4 inch equals 1 foot?
- Show me the path from the door to the bed. Where is it narrowest?
- Why does a bed need one open side? Who is that side for?
- Your peer wants the desk under the window. What does the client gain and lose?
Notes: The desk-under-the-window answer: more light, but the window may not open and a wheelchair user may not reach the latch. Take both sides.

Slide 19: Revise, 12 minutes

- Fix every failed path
- Fix every checklist no you agree with
- Glue or tape the final pieces down
- On the back of the plan: two changes you made and why, one suggestion you rejected and why Notes: Go straight to the plans with a circled path. The push is always the same: you have a 28-inch path and a 36-inch rule. What moves?

Slide 20: Before and after

- Two pairs hold up the plan
- The class names what moved and why
- Three minutes Notes: Pick one plan where a path was fixed and one where a suggestion was rejected for a good reason. Rejecting well is a skill.

Slide 21: Exit card

- One path in my plan is ___ inches wide, which passes because ___ Notes: Sort the cards into pass and fail on the math. “It passes because it looks big enough” is a fail; those students get a two-minute check at the start of Lesson 4.13. Preview: tomorrow every piece on this plan has to be paid for.